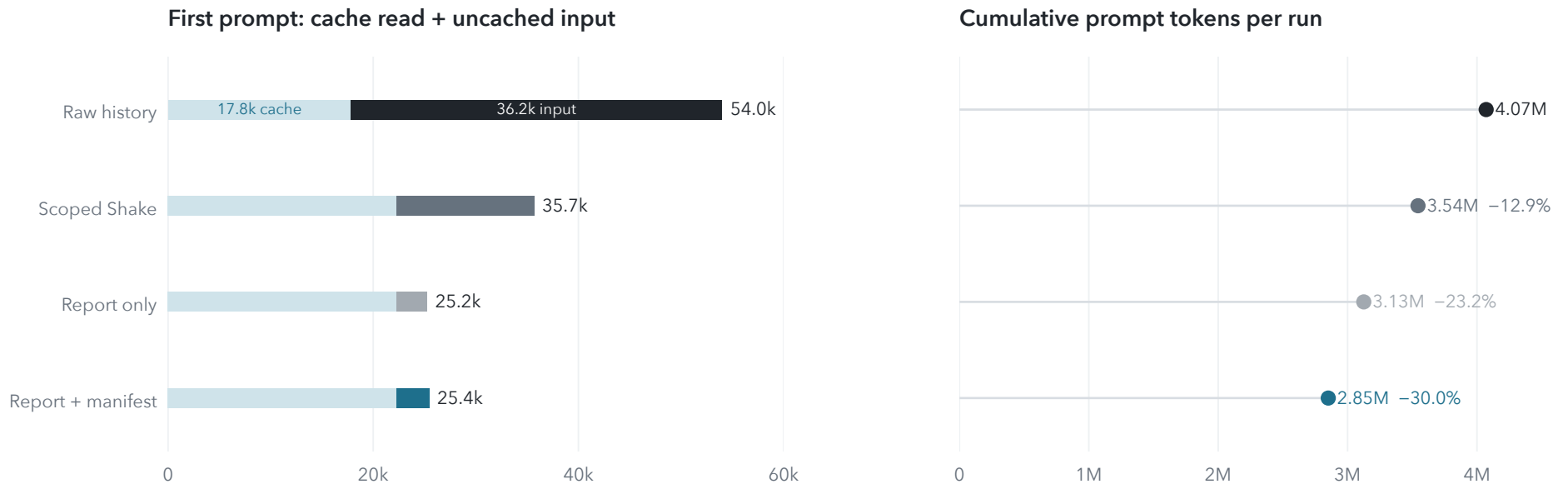


Cache reads were ubiquitous—and still the largest cost

1

20 valid runs · 1,120 provider turns · absolute token accounting, not a ratio-only view



CACHE AND COST TOTALS



99.82%

cache-read incidence

1,118 of 1,120 turns had cacheRead > 0.

66.1M

cached prompt tokens

Cached tokens were 97.2% of prompt volume.

\$0.50

cold-cache premium

Two cold turns; no quality inference.

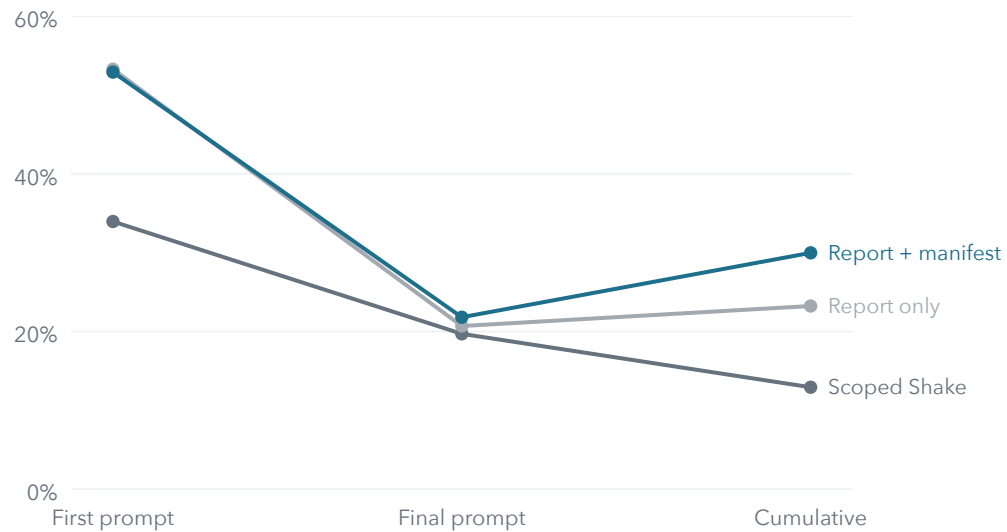
Cached tokens were discounted—not free—and still consumed context capacity on every turn.

Initial savings decayed; run stochasticity dominated quality

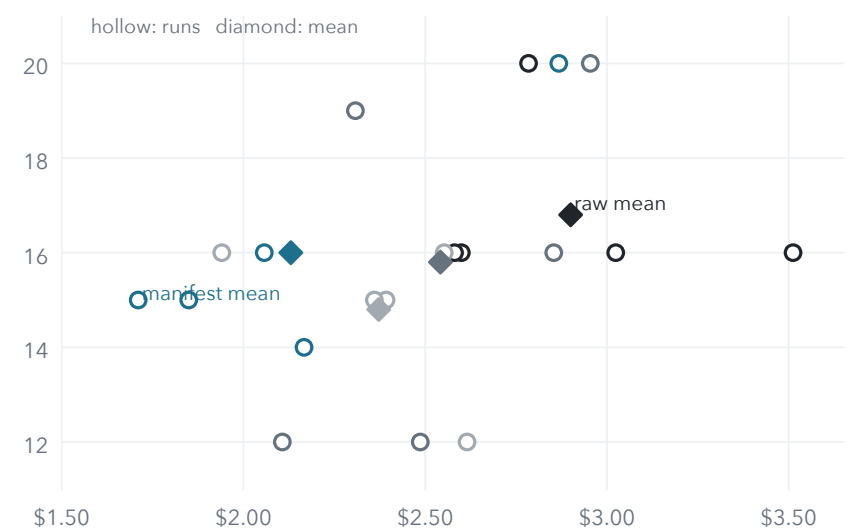
2

Lifecycle context reduction, continuation economics, and quality-cost overlap

Reduction versus raw over the trajectory



Run-level quality versus continuation cost



ONE-SHOT END-TO-END ECONOMICS

Raw history

\$2.90

8.15m

Scoped Shake

\$2.88

10.30m

seal: +\$0.34, +1.43m

Report only

\$2.66

9.51m

seal: +\$0.28, +1.80m

Report + manifest

\$2.41

8.63m

seal: +\$0.28, +1.80m

9.0%

score variance explained by treatment

ANOVA $p=.669$; treatment means are not resolved.

-30.0%

manifest cumulative prompt

Initial: 52.9%. Reconstruction reduced lifecycle savings.

$K \geq 2$

manifest time break-even

Two continuations amortize 1.80m seal latency.

One continuation task, four isolated context treatments

What the corrected experiment measured—and what it could not measure

COMMON START

Verified S1 checkpoint: transient Bloomberg request workflow implemented; CLI integration intentionally unfinished.

Identical filesystem seed, todo state, continuation prompt, model, tools, evaluator, and canonical cwd.

RAW	SCOPED SHAKE	REPORT ONLY	REPORT + MANIFEST
120 messages · 54.0k	120 messages · 35.7k	6 messages · 25.2k	7 messages · 25.4k
Untreated user, assistant, and tool trajectory.	51 tool payloads elided; chronology retained.	Semantic handoff without runtime provenance.	The same report plus 198 provenance tokens.
same CLI continuation task		21 hidden contracts	cost · time · tokens

WHAT IT MEASURES	BLIND SPOTS
- Continuation behavior conditional on one shared implementation checkpoint. - Repository grounding, editing, repair, and exact CLI output contracts. - First/final/cumulative context, provider cost, tools, and hidden behavior.	- One Python task, repository, evaluator, model, provider, and price regime. - One generated semantic seal and one Shake derivation reused five times. - Warm shared prefixes; no TTFT, live Bloomberg, external effects, or multi-agent work.

HIGHEST-LEVERAGE METHOD UPGRADES

Nested seal replication	More tasks before more repeats	Pre-register and instrument
Several reports per task and continuations per report separate handoff variance from run variance. Provider turns are nested observations; the experimental unit is a complete continuation run.	Block debugging, migration, feature, and research tasks; estimate task effects.	Predefine quality floors, cache/TTFT, invalidation, and one-shot economics.

Use the result to narrow the product—and broaden the research

Product modes, prioritized research, and an automated evidence pipeline

PRODUCT MODES

SAFETY DEFAULT

Raw / keep

Strongest observed floor. Prefer when exact trajectory detail is load-bearing or cache is cheap and window pressure is low.

CONTEXT PRESSURE

Report + manifest

Best mean quality-cost frontier. Prefer for long remaining work, handoff, or branch fan-out; disclose seal overhead.

CHRONOLOGY REQUIRED

Scoped Shake

Retains all message structure. Do not market as the general efficiency mode; it did not beat the semantic candidate here.

PRIORITIZED RESEARCH

P0	Cold/warm cache probes	4 conditions × 2 cache states × 3 short probes	Calibrate cache continuity, TTFT, and true first-turn economics.
P1	Multi-task external validity	4 task types × 4 treatments × 3 replicates	Estimate task-level effects; pre-register quality floor and end-to-end cost.
P1	Manifest ablation	Report, minimal manifest, evidence manifest, full manifest	Identify which approximately 198 structured tokens reduce rediscovery.
P1	Verification-floor gate	Raw/manifest × normal/required evidence × 5	Test whether mandatory repair loops remove low-cost, low-score tails.
P2	Lifecycle and retrieval	Single seal, re-seal, threshold re-seal, artifact retrieval	Measure context regrowth without compounding semantic omission.
P2	Fan-out amortization	One seal feeding K=1, 2, 4 continuations	Measure actual branch cache behavior and seal break-even.

AUTOMATE THE EVIDENCE events > validity gates > typed tables > versioned metrics > pre-registered statistics > reports and dashboards. Never auto-seal from todo completion alone.